

Statistical Foundations of Measures of Association

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Introduction

Measures of association

Measures of association are used to answer a recurring question across empirical sciences:

Are two quantities genuinely related, or related by chance?

- First conceptualised by **Galton** – size of pea seeds across generations
- Later formalised by **Pearson** as “correlation”
- Followed by **Spearman** – marks of students across subjects

Classical measures of association

Let $\{(X_i, Y_i)\}_{i=1}^n$ be n i.i.d observations from (X, Y) .

Pearson's correlation

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

For finite sample

$$r_{XY} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

Spearman's rank correlation

$$r_S = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

where, $d_i = R[X_i] - R[Y_i]$

where $R[X_i], R[Y_i]$ are the ranks of X_i among X and Y_i among Y respectively.

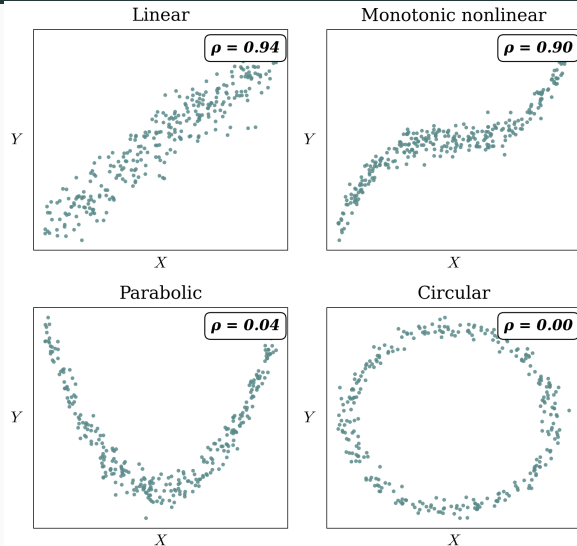
Properties

Pearson's correlation has:

- Statistical guarantees.
- Clear interpretation.

But:

- Detect only linear relationships
- Built for scalar variables.



Broad research aim

To develop flexible measures of association equipped with the statistical theory needed for inference.

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- Suitable to test most types of association
- No distributional assumptions on how data is generated

To develop flexible measures of association equipped with the statistical theory needed for inference.

- Knowing distribution of the finite sample measure
- Usable for applications – variable selection, hypothesis testing

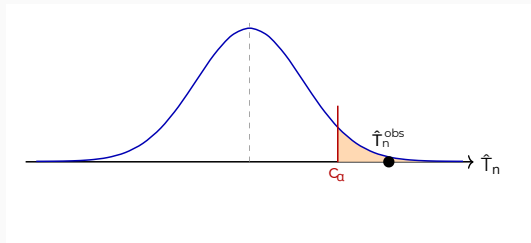
Significance Testing

For a measure $\hat{T}_n$ we have

- $H_0: X \perp Y$
- $H_A: X \not\perp Y$
- For significance level α

$$p = \mathbb{P}_{H_0}(\hat{T}_n \geq \hat{T}_n^{obs})$$

- Reject H_0 when $p \leq \alpha$



Modern measures — more flexible, mixed on theory

Family	Examples	Properties
Linear	RV coefficient ¹ , Canonical Correlation ²	Linear association, Full statistical theory, Multivariate
Kernel-based	HSIC ³ , Kernel Canonical Correlation ⁴	Non-linear association, Full statistical theory under null, Multivariate
Rank-based	Friedman-Rafsky coefficient ⁵ , Chatterjee's coefficient ⁶ , Azadkia–Chatterjee coefficient ⁷	Directional, Non-linear association, Statistical theory not fully developed, Multivariate + Partially multivariate

¹Escoufier (1973); ²Hotelling (1992); ³Gretton et al. (2007); ⁴Gretton et al. (2005); ⁵Friedman & Rafsky (1983); ⁶Chatterjee (2021); ⁷Azadkia & Chatterjee (2021).

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Ongoing work

Information Imbalance

Information Imbalance¹ (II) is a rank-based multivariate, distribution free measure of association.

Definition

Let $(X_1, Y_1), \dots, (X_n, Y_n) \stackrel{\text{iid}}{\sim} f_{X,Y}$ on $\mathbb{R}^d \times \mathbb{R}^p$.

$$\hat{\Pi}_n(X \rightarrow Y) = \frac{2}{n^2} \sum_{i=1}^n R_i^Y(N_X(i)),$$

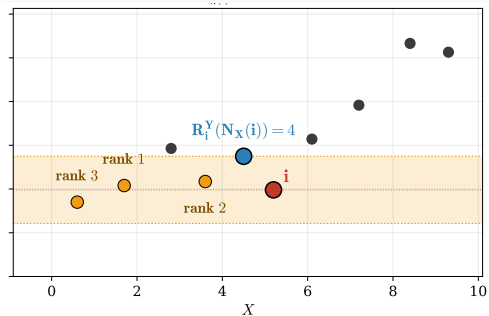
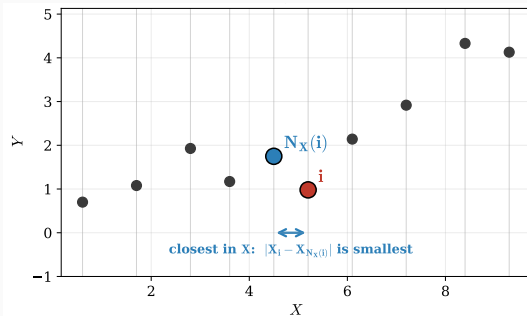
where $N_X(i)$ is the index of nearest neighbour of X_i and $R_i^Y(N_X(i))$ is the rank of $Y_{N_X(i)}$ among neighbours of Y_i .

$\hat{\Pi}_n(X \rightarrow Y) < \hat{\Pi}_n(Y \rightarrow X) \implies X$ contains more information about Y than vice versa.

Idea: If points close in X are also close in Y , then X can provide information about Y .

¹ Glielmo, A., Zeni, C., Cheng, B., Csányi, G. and Laio, A., 2022. Ranking the information content of distance measures.

Rank Computation

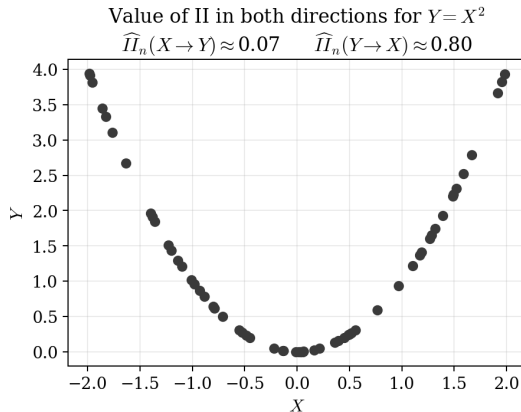


Properties of the estimator

- It is asymmetric in nature,
i.e.,
 $\hat{\Pi}_n(X \rightarrow Y) \neq \hat{\Pi}_n(Y \rightarrow X)$

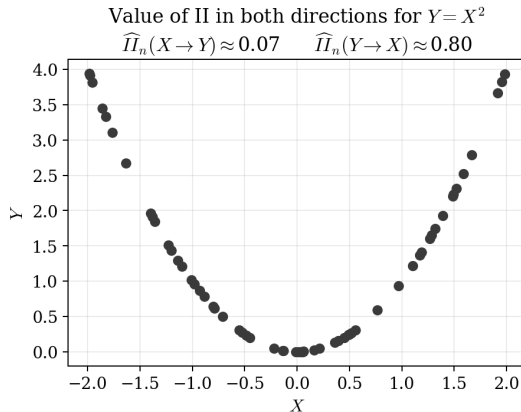
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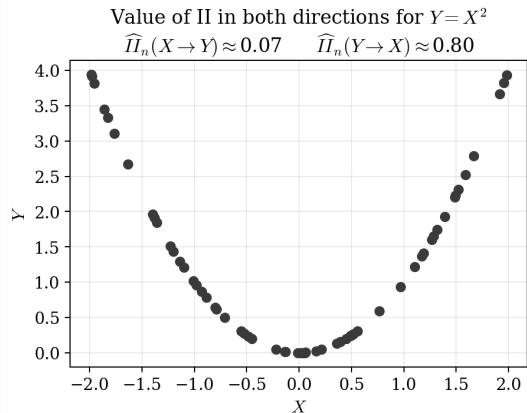
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- $\hat{\Pi}_n(X \rightarrow Y) \approx 0$ when Y is
a function of X .



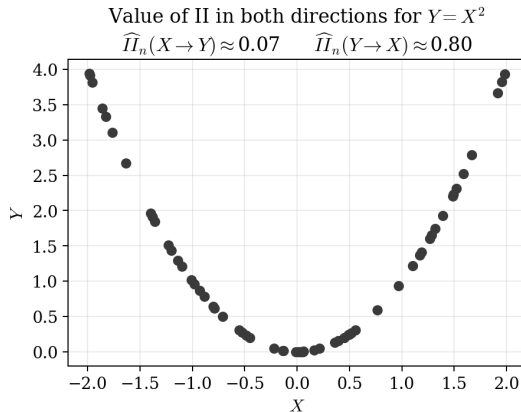
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Without a population limit and knowledge of sampling distribution it is only a descriptive tool, not an inferential one.

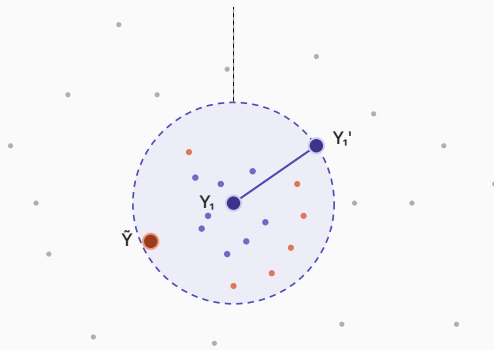
Result 1 — Asymptotic limit

Theorem

For any continuous joint density function $f_{X,Y}$, as $n \rightarrow \infty$

$$\hat{\Pi}_n(X \rightarrow Y) \xrightarrow{a.s.} \Pi(X \rightarrow Y) := 2 \mathbb{P}(\|Y_1 - \tilde{Y}\| \leq \|Y_1 - Y'_1\|),$$

where $\tilde{Y} \sim f_Y$, $X_1 \sim f_X$, and $Y_1, Y'_1 \stackrel{iid}{\sim} f_{Y|X=X_1}$.



What does the limit convey?

Theorem

For any continuous joint density function $f_{X,Y}$, as $n \rightarrow \infty$

$$\hat{\Pi}_n(X \rightarrow Y) \xrightarrow{a.s.} \Pi(X \rightarrow Y) := 2\mathbb{P}\left(\|Y_1 - \tilde{Y}\| \leq \|Y_1 - Y'_1\|\right),$$

where $\tilde{Y} \sim f_Y$, $X_1 \sim f_X$, and $Y_1, Y'_1 \stackrel{iid}{\sim} f_{Y|X=X_1}$.

Two extreme cases: $\Pi(X \rightarrow Y) = 0$ if $Y = g(X)$ and $\Pi(X \rightarrow Y) = 1$ if $X \perp\!\!\!\perp Y$.

Rate of convergence to limit: $\mathcal{O}\left(n^{-\min\{1/2, 1/d\}}(\log n)^{d+1+\beta}\right)$

Result 2 — Central Limit Theorem

Theorem

For any continuous joint density function $f_{X,Y}$ such that Y is not a.s. a measurable function of X ,

$$\frac{\hat{\Pi}_n - \mathbb{E}[\hat{\Pi}_n]}{\sqrt{\text{Var}(\hat{\Pi}_n)}} \xrightarrow{d} \mathcal{N}(0, 1).$$

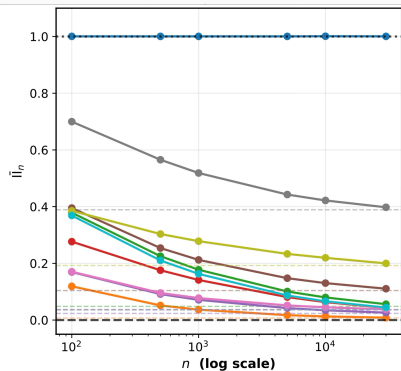
The proof strategy takes inspiration from the line of research on establishing guarantees for Chatterjee's and Azadkia-Chatterjee's coefficient.

Simulation results

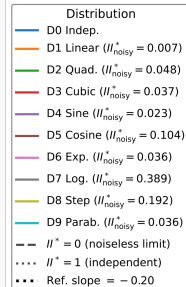
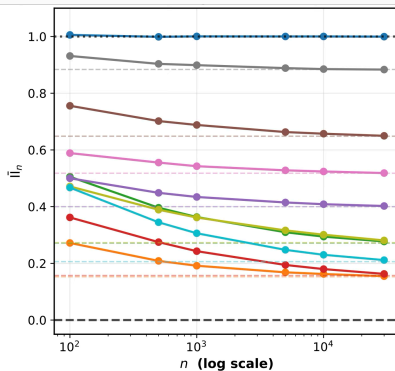
Empirical verification — consistency

$$Y=f(X) + \text{noise}$$

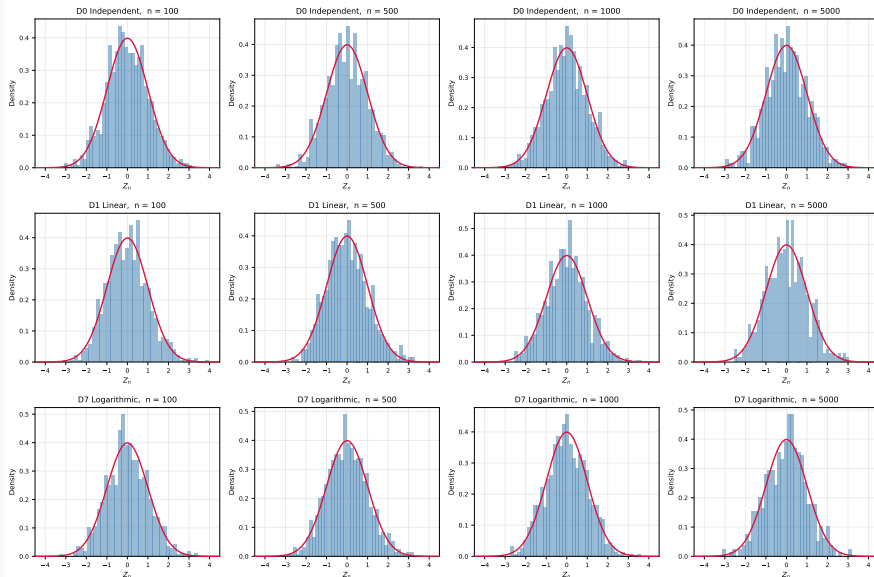
Convergence (low noise)



Convergence (high noise)



Empirical verification — asymptotic normality



Research Plan

What's next — Closing the theory

The CLT is not yet usable for testing as the asymptotic variance has no closed form.

Next questions:

- Estimation of variance
- Estimation of bias : $\mathbb{E}[\hat{\Pi}_n] - \Pi$
- Power analysis

What's next — Beyond II

Methodological extensions:

- Measures of **conditional** association
- Measures of association for **discrete variables**.

Applications

- **Causal discovery** — measures of conditional association + directionality.
- **Fairness in ML** — fairness criteria are statements about (conditional) dependence.
- **High-dimensional variable selection** — nonparametric, model-free, with consistency guarantees.

Theoretical foundations

Asymptotic limit

Central limit theorem

Closed-form variance

Conditional & multivariate extensions

Flexible measures of association

with statistical guarantees

Applications

Causal discovery

Fairness in ML

High-dimensional
variable selection

Thank you

Why Pearson fails on nonlinear data?

Pearson is the cosine of the angle between X and Y as vectors:

$$\rho = \frac{\langle X - \bar{X}, Y - \bar{Y} \rangle}{\|X - \bar{X}\| \cdot \|Y - \bar{Y}\|}.$$

By construction it measures only the signed linear component.

- **Parabola:** positive and negative contributions cancel by symmetry.
- **Circle:** every quadrant cancels.

Assumptions in full

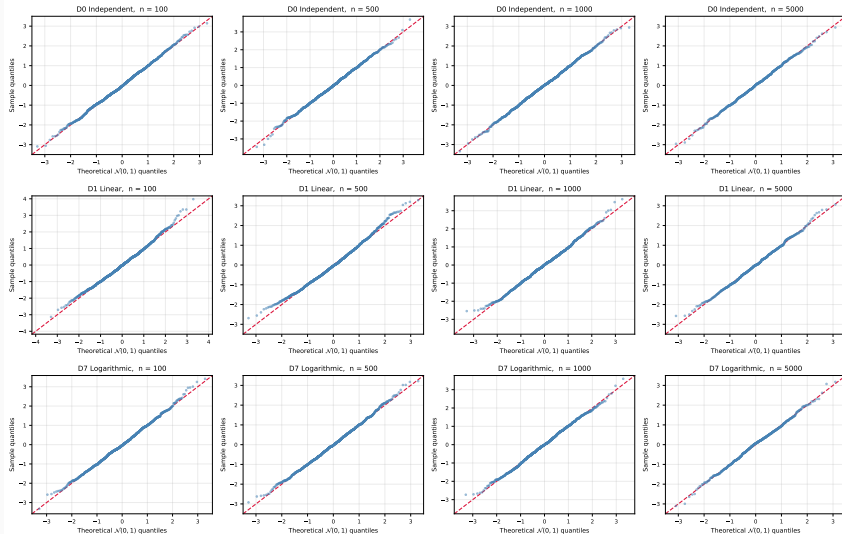
A1 $F_{Y|X}$ satisfies local Lipschitz with polynomial growth exponent $\beta \geq 0$.

$$|\mathbb{P}(Y \geq t \mid X = x) - \mathbb{P}(Y \geq t \mid X = x')| \leq C(1 + \|x\|^\beta + \|x'\|^\beta)(\|x - x'\|)$$

A2 Marginal of X has sub-exponential tails .

$$\mathbb{P}(X \geq t) \leq C_1 e^{-C_2 t}$$

Empirical verification — asymptotic normality



Proof Sketch – CLT

We establish the chain of approximations

$$\lim_{n \rightarrow \infty} n\text{Var}(II - \check{II}) = 0$$

$$\lim_{n \rightarrow \infty} n\text{Var}(\check{II} - II^*) = 0$$

$$(II^* - \mathbb{E}[II^*])/\sqrt{\text{Var}(II^*)} \rightarrow N(0, 1)$$

- KW distance is defined as

$$\mathcal{W}(\mu, \nu) = \sup\left\{\left|\int h d\mu - \int h d\nu\right| : \|h\|_{Lip} \leq 1\right\}$$